HTTP Methods – API Testing Using Postman

GET, POST, PUT, PATCH and DELETE — Beginner-Friendly Practical Guide

1. What are HTTP Methods?

HTTP methods tell a web server what operation an API client wants to perform on a resource. In REST APIs, the five commonly used methods are GET, POST, PUT, PATCH and DELETE.

Method	Purpose	Simple Meaning
GET	Retrieve data	Read / view data
POST	Create data	Add a new record
PUT	Replace/update data	Update the complete record
PATCH	Partially update data	Update selected fields
DELETE	Remove data	Delete a record

2. What is Postman?

Postman is a tool used to send API requests and inspect API responses. You can select an HTTP method, enter the API URL, add query parameters, headers and a JSON request body, then click Send. The response normally contains a status code, headers and response data.

Basic flow: Client/Postman → HTTP Request → API/Server → HTTP Response → Postman

3. Example API Used in This Manual

For practice, assume a local Express.js API is running at **http://localhost:5000/api/users**. The examples use a user resource.

GET – Fetch Data

URL: GET http://localhost:5000/api/users

Purpose: Get the list of users. Usually no request body is required.

```
{
"users": [
{"id": 1, "name": "Salil", "email": "salil@example.com"}
]
}
```

POST – Create Data

URL: POST http://localhost:5000/api/users

Purpose: Create a new user. In Postman choose Body → raw → JSON.

```
{
"name": "Rahul",
"email": "rahul@example.com",
"password": "123456"
}
```

PUT – Update/Replace Data

URL: PUT http://localhost:5000/api/users/5

Purpose: Replace/update the complete user record represented by ID 5

```
{
"name": "Amit Kumar",
"email": "amit@example.com",
"password": "newpassword"
}
```

PATCH – Partial Update

URL: PATCH http://localhost:5000/api/users/5

Purpose: Change only selected fields. Other fields remain unchanged

```
{
"email": "newemail@example.com"
}
```

DELETE – Remove Data

URL: DELETE http://localhost:5000/api/users/5

Purpose: Delete the user whose ID is 5. Usually no request body is required

```
No request body
```

4. How to Test an API in Postman – Step by Step

Step 1 – Open Postman

Open Postman and create a new HTTP request.

Step 2 – Select Method

Select GET, POST, PUT, PATCH or DELETE from the method dropdown.

Step 3 – Enter URL

Enter the API endpoint, for example http://localhost:5000/api/users.

Step 4 – Add Parameters

For GET requests, query parameters can be added in the Params tab, such as ?page=1.

Step 5 – Add Headers

For JSON requests, commonly use Content-Type: application/json. Authentication headers may also be required.

Step 6 – Add Body

For POST, PUT and PATCH, select Body → raw → JSON and enter the JSON data.

Step 7 – Click Send

Postman sends the request to the server.

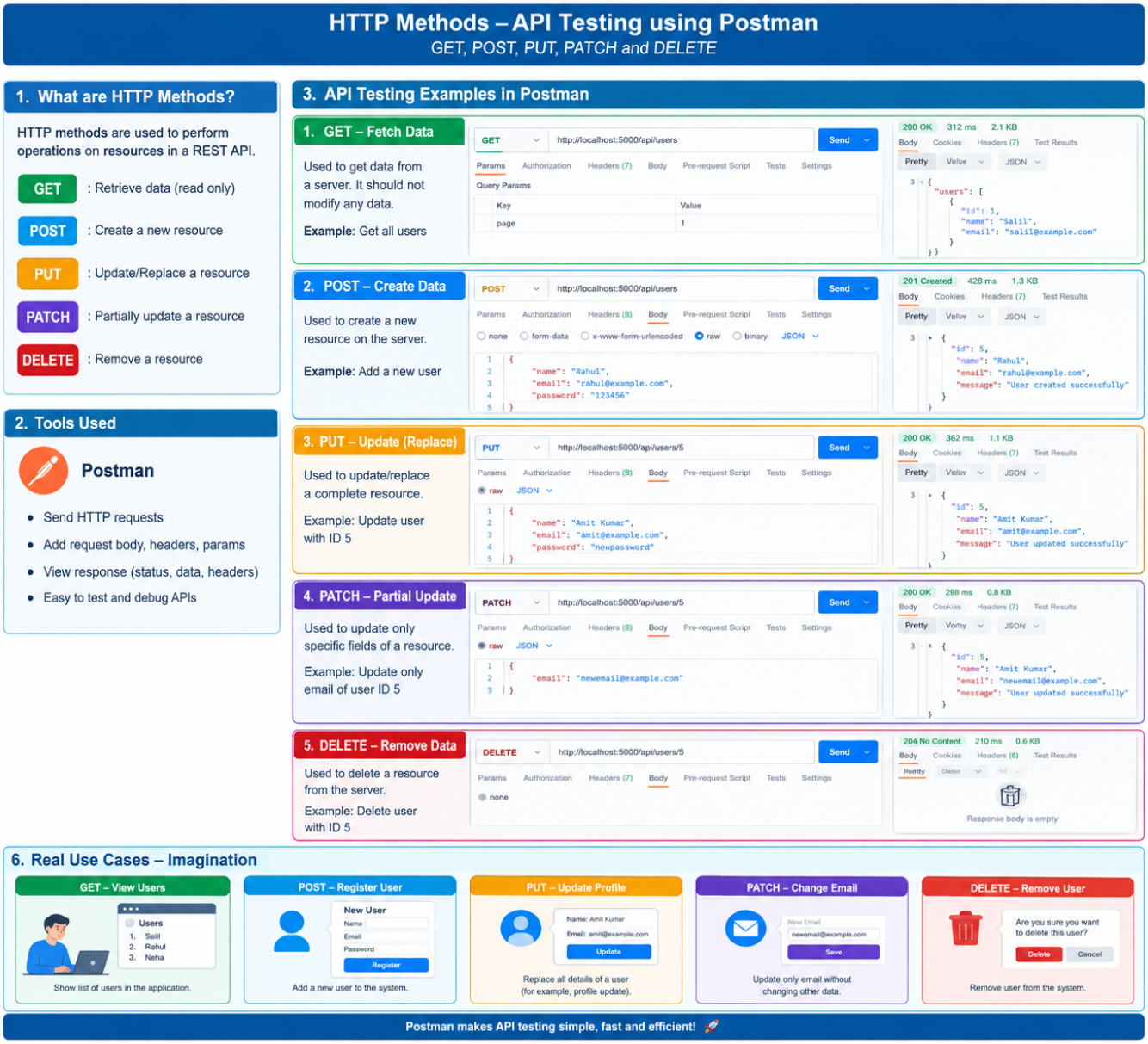
Step 8 – Check Response

Check the HTTP status code, response body, response headers and response time.

5. Common HTTP Status Codes

Code	Meaning	Typical Use
200 OK	Request successful	GET, PUT, PATCH
201 Created	Resource created	POST
204 No Content	Successful request with no response body	DELETE
400 Bad Request	Invalid request data	Validation error
401 Unauthorized	Authentication required/failed	Protected API
404 Not Found	Resource/route not found	Wrong URL or ID
500 Internal Server Error	Server-side error	Backend exception

6. Visual Imagination / Real-World Use



Remember: GET = View, POST = Create, PUT = Replace/Update, PATCH = Partial Update, DELETE = Remove.

7. Quick Postman Practice Table

Task	Method	Example Endpoint
Show all users	GET	/api/users
Show one user	GET	/api/users/5
Add a user	POST	/api/users
Update complete user	PUT	/api/users/5
Change only email	PATCH	/api/users/5
Delete a user	DELETE	/api/users/5